

Democracy in Darkness

Metric Voting Under Worst-Case Misinformation

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Motivation & Background

Modeling Ballots and Voting

- Each voter ranks the candidates from most to least preferred
- $A \succ_v B$ means that voter v ranks A above B
- An **ordinal ballot** records the order of preference ranking

$$A \succ_v B \succ_v C$$

- A **voting rule** combines the voters' ballots and selects a winner w

Motivation & Background

Single-Choice and Ranked Ballots

Single-choice ballot

Ranked ballots

Choose one candidate

Rank the candidates

✓ A

1. A

B

2. B

C

3. C

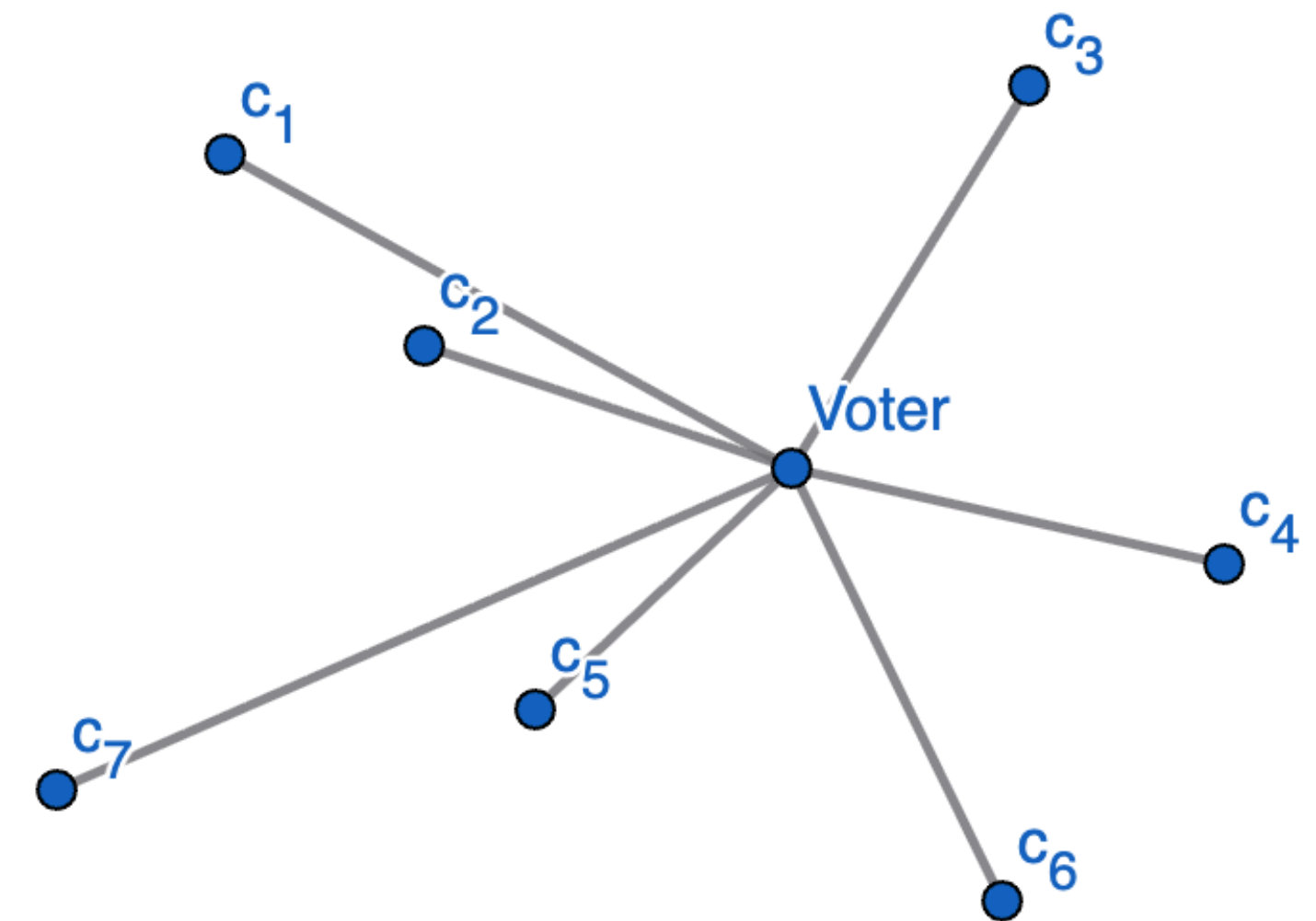
Used by Plurality

Used by rules such as
Borda and STV

Motivation & Background

Modeling Preferences with Distances

- V = voters, C = candidates
- d is a metric on $V \cup C$
- $d(v, c)$ is voter v 's cost if candidate c wins
- Smaller distance means lower cost



* This is a simplified model of preferences, not a claim about literal political geography.

Motivation & Background

Standard Metric Distortion

- In the standard model,

$$c \succ_v c' \quad \Rightarrow \quad d(v, c) \leq d(v, c')$$

- The metric satisfies the triangle inequality

$$d(v, c) \leq d(v, c') + d(v', c') + d(v', c)$$

- Select the best candidate based on ordinal ballots submitted by voters

Evaluate winner by $\sum_{v \in V} d(v, w)$

Motivation & Background

Standard Metric Distortion

- Let social cost of some candidate c be defined as

$$SC(c) = \sum_{v \in V} d(v, c)$$

- w is the winner selected by the voting rule
- c^* is the socially optimal candidate: the candidate with minimum social cost

- Distortion of a voting rule is defined as $\max_d \frac{SC(w)}{SC(c^*)}$

- m is the number of candidates

Motivation & Background

Ordinal ballots reveal order, not strength of preference

- But preference rankings only tell us which candidate a voter prefers—not by **how much**
- As a result, the same observed voting outcome can represent very different underlying strengths of preference

Motivation & Background

Numerical Example

$$B \succ_{v_1} A$$

$$A \succ_{v_2} B$$

Winner: B

Voter	Distance to Candidate A	Distance to Candidate B
1	51	50
2	0	101

$$\text{distortion on this instance} = \frac{SC(w)}{SC(c^*)} = \frac{50 + 101}{51 + 0} \sim 3$$

Motivation & Background

Personal Experience and National Perceptions can Diverge

Own household finances

54%

rated or leaned “good”



The national economy

73%

rated or leaned “poor”



same survey
different vantage points

National judgments need not mirror personal experience.

AP–NORC, October 2023 · N = 1,163 U.S. adults

Motivation & Background

- **Standard assumption:** every reported comparison agrees with the underlying distances
- **In practice:** voters may be poorly informed about candidates and can make mistakes, especially when candidates are similar distance from the voter

Which rules continue to stay robust to poorly informed voters?

Our Approach

α -distortion

- We introduce a parameter $\alpha \geq 1$ which represents the potential worst-case misinformation of a voter
- Relax the assumption:

$$d(v, c) \leq \alpha d(v, c') \quad \text{if} \quad c \succ_v c'$$

- This is a natural generalization of the original metric distortion model

Related Work

Current Models

- **Probabilistic / Noise Models:**
 - Goyal and Sarmasakar 2025 modeled metric distortion with stochastic errors
 - Caragiannis and Micha, 2019 modeled liquid democracy with probabilistic rankings and a common preferred candidate
- Probabilistic models impose a distribution over rankings
- We study a worst case misinformation model

Result Overview

α -distortion

Mechanism	Metric Distortion		α -Distortion	
	Lower Bound	Upper Bound	Lower Bound	Upper Bound
PLURALITY	$2m - 1$	$2m - 1$	$m(\alpha + 1) - 1$	$m(\alpha + 1) - 1$
BORDA	$2m - 1$	$2m - 1$	$\alpha m + 1$	$2\alpha m$
PLURALITY VETO	3	3	$\alpha^2 + 2\alpha$	$\alpha^2 + 2\alpha$
COPELAND	5	5	$2\alpha^2 + 2\alpha + 1$	$2\alpha^2 + 2\alpha + 1$
STV	$\Omega\left(\frac{\log m}{\log \log m}\right)$	$O(\log m)$	$\min(\alpha, 2)^{\Omega(\sqrt{\log m})}$	$O(\max(\alpha m, \alpha^2))$
SINGLE-ELIMINATION TOURNAMENT	—	—	$2\alpha^{\ell-1} + 1$	$4\ell\alpha^{\ell}$
Optimal Rule	3	3	$\Omega(\min(\alpha m, \alpha^2))$	$O(\max(\alpha m, \alpha^2))$

The Analytical Tool: Dual Flow Framework

Transforming into Linear Programming Problem

- **The Baseline (Kempe 2020):**
 - Expresses metric distortion bounds as a linear programming problem
 - In the standard framework, finding a upper bound on metric distortion is equivalent to constructing a valid flow on an election graph

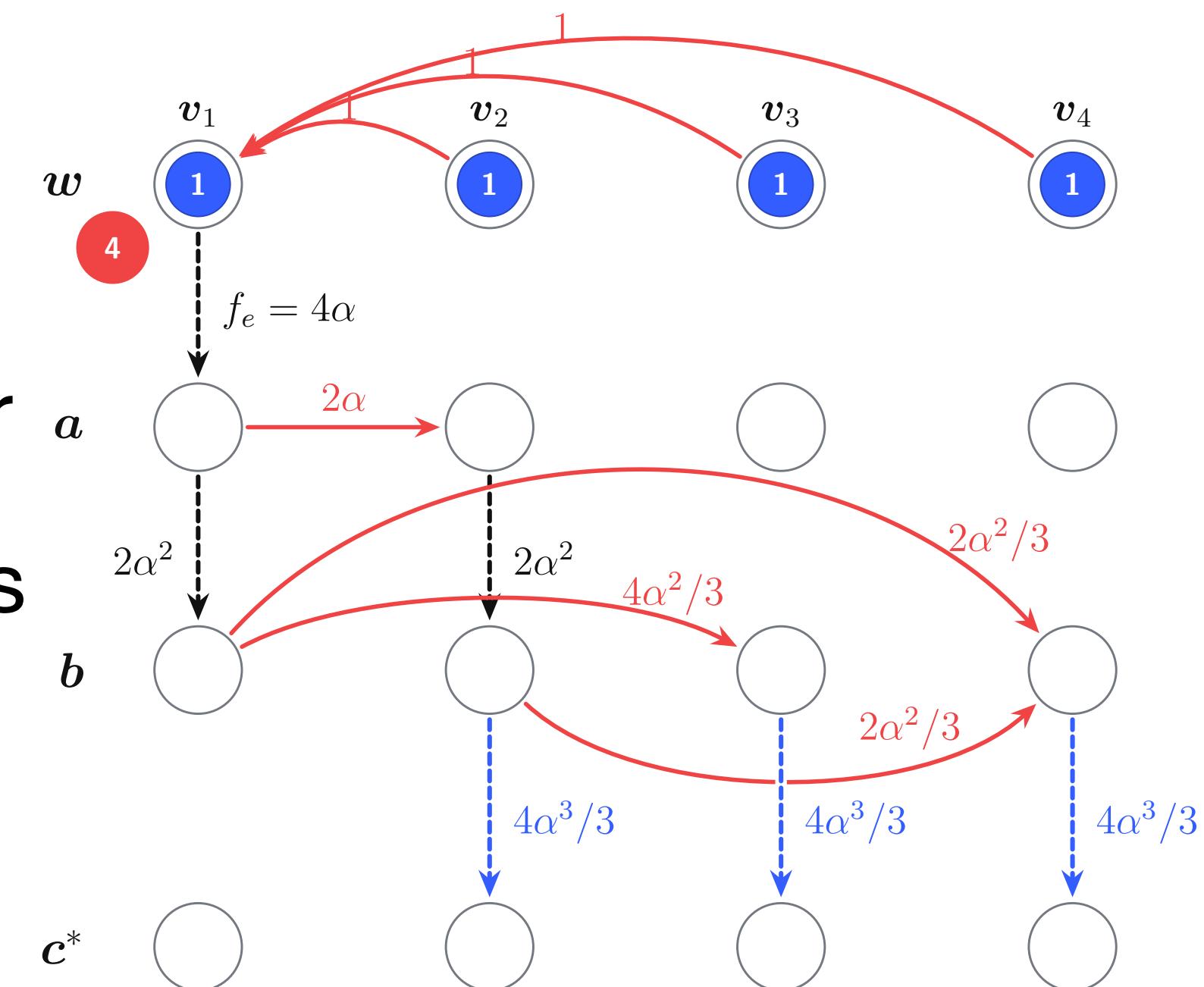
- **Our Generalization:**

$$\begin{array}{ll} \text{Maximize} & \sum_v d_{v,w} \\ \text{subject to} & d_{v,c} \leq d_{v',c} + d_{v',c'} + d_{v,c'} \quad \text{for all } c \neq c', v \neq v' \text{ (Triangle Inequality)} \\ & d_{v,c} \leq \alpha \cdot d_{v,c'} \quad \text{for all } c, c', v \text{ such that } c \succ_v c' \text{ (consistency)} \\ & \sum_v d_{v,c^*} = 1 \quad \text{(normalization)} \\ & \sum_v d_{v,c} \geq 1 \quad \text{for all } c \text{ (optimality of } c^*) \\ & d_{v,c} \geq 0 \quad \text{for all } c, v. \end{array}$$

The Analytical Tool: Dual Flow Framework

Flow Construction

- Vertices are (v, c) pairs, candidates are rows and voters are columns
- Horizontal edges connects vertices of the same candidate
- Vertical (preference) edges connects vertices of the same voter
 - Edges are directed from preferred to dispreferred candidates
 - Flow blows up by a factor of α
- Each voter has flow 1 starting from (v, w) and eventually getting absorbed at any c^* vertex
- Costs of the flow are incurred at each horizontal edge at both voters and when the flow eventually gets absorbed



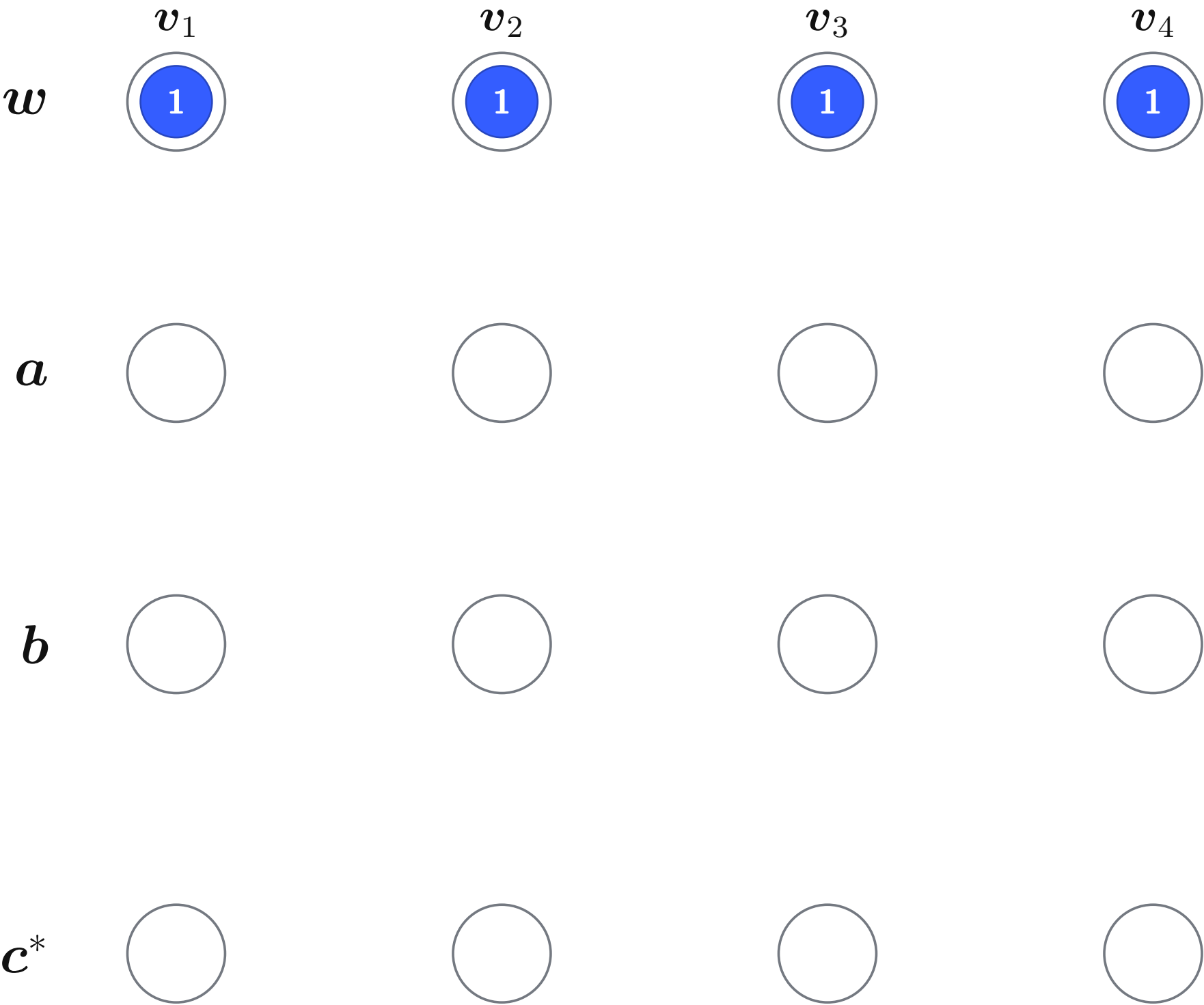
The Analytical Tool: Dual Flow Framework

Flow Construction

Theorem. $\text{distortion} \leq \max_{v \in V} \sum_{c \in C} \text{cost}(v, c)$

The Analytical Tool: Dual Flow Framework

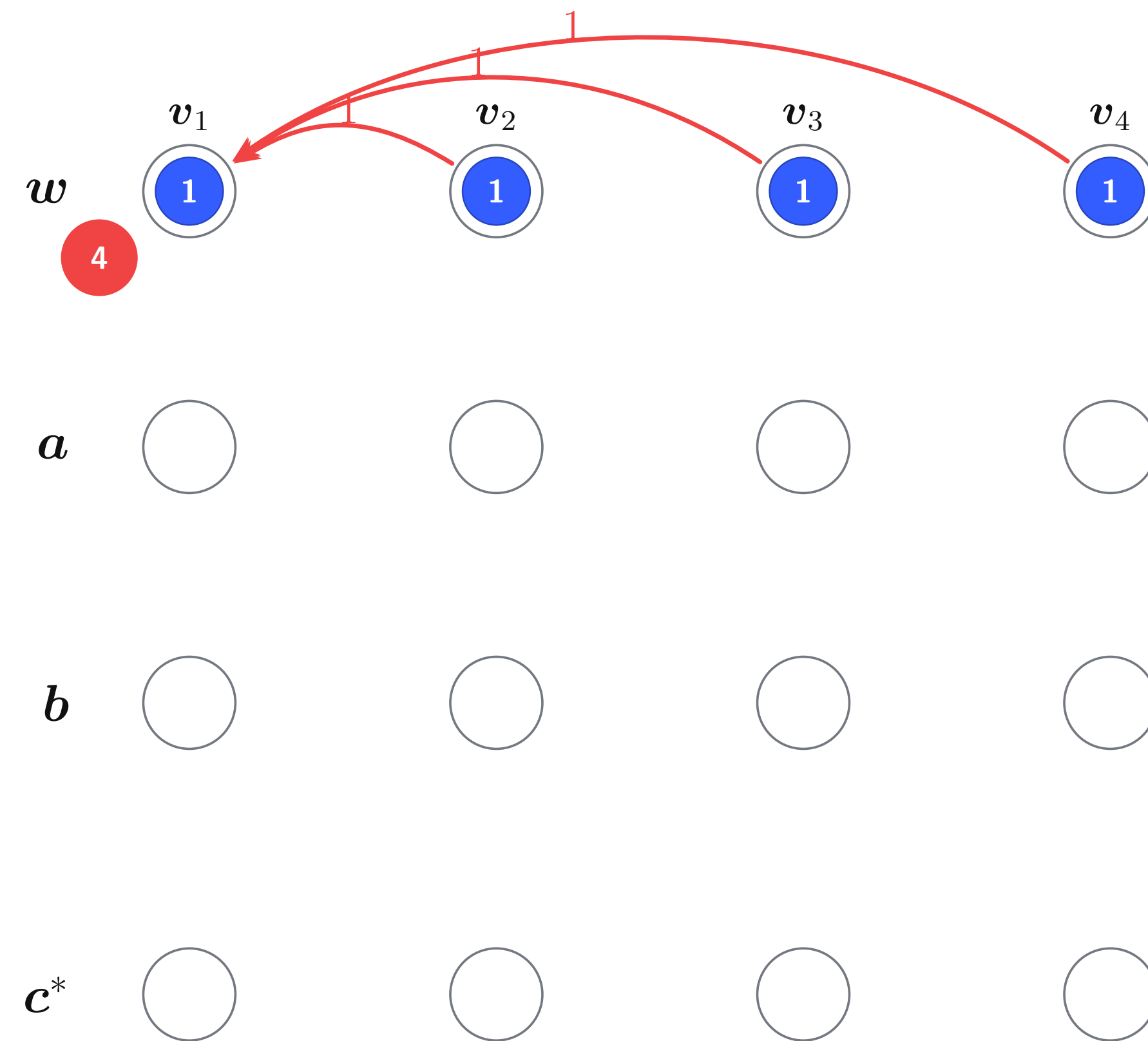
Graph construction and visual mechanics



The Analytical Tool: Dual Flow Framework

Graph construction and visual mechanics

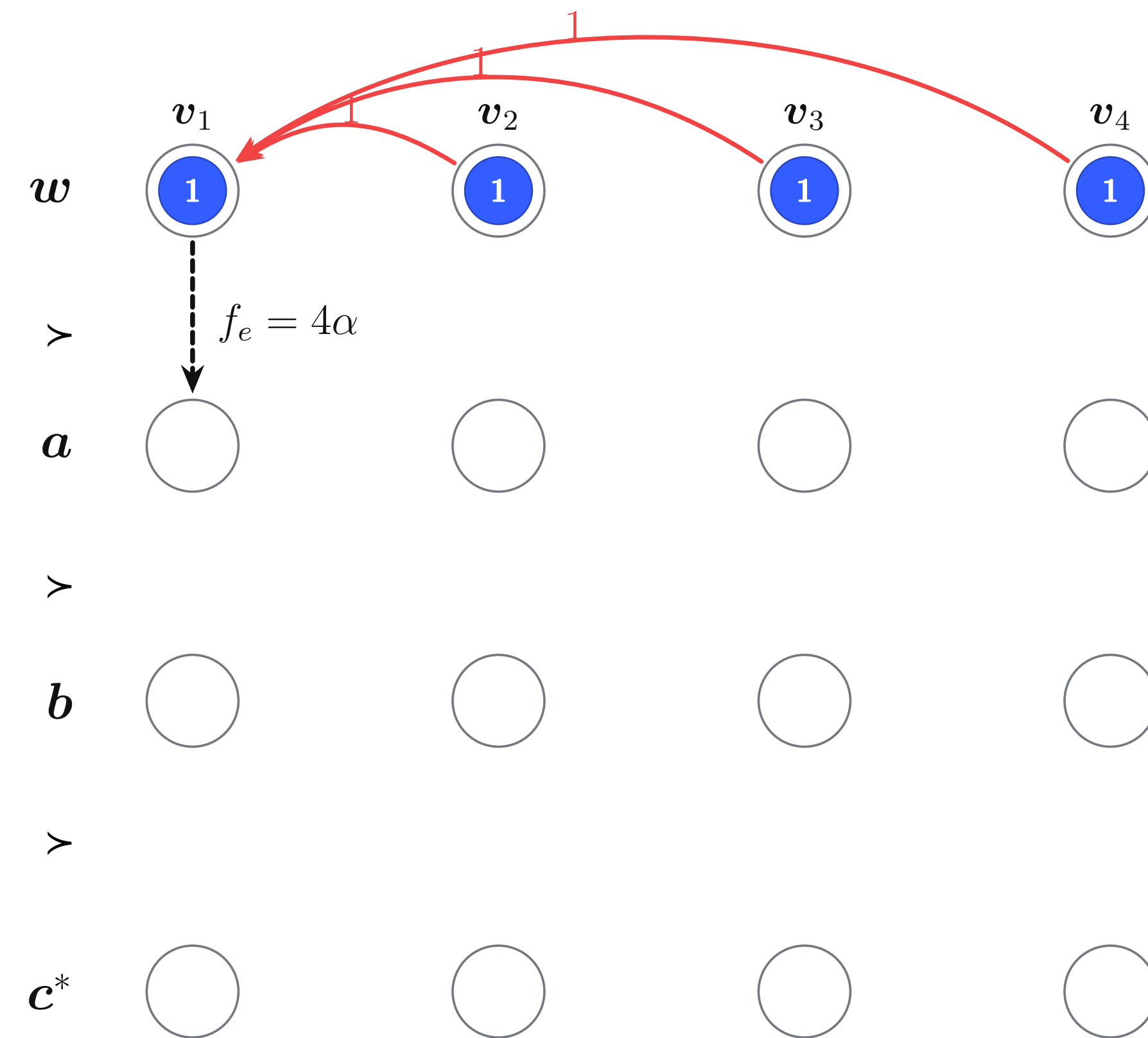
- *Triangle-inequality dual variables* are the horizontal sideways edges



The Analytical Tool: Dual Flow Framework

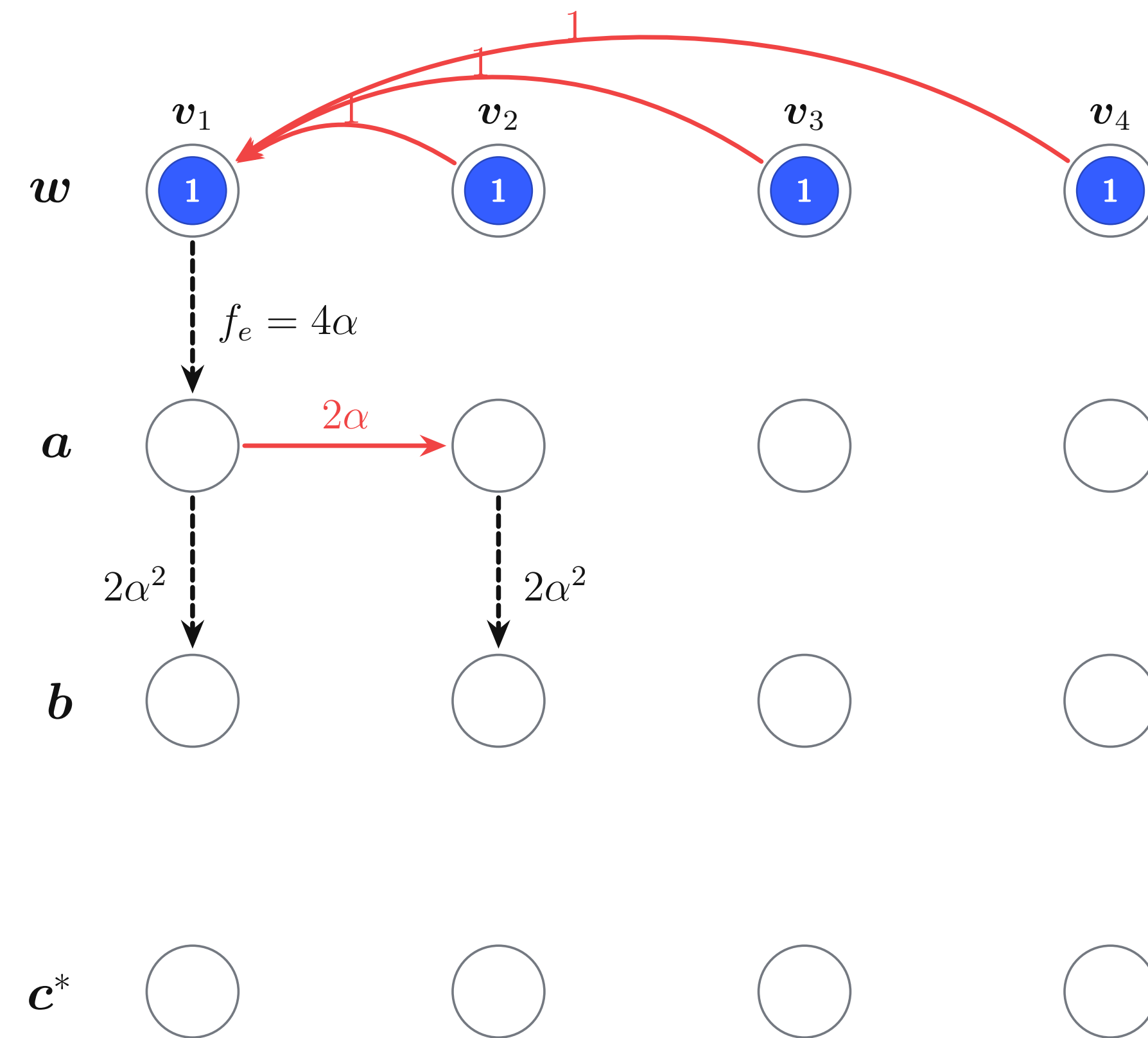
Graph construction and visual mechanics

- *Consistency dual variables* are the vertical preference edges



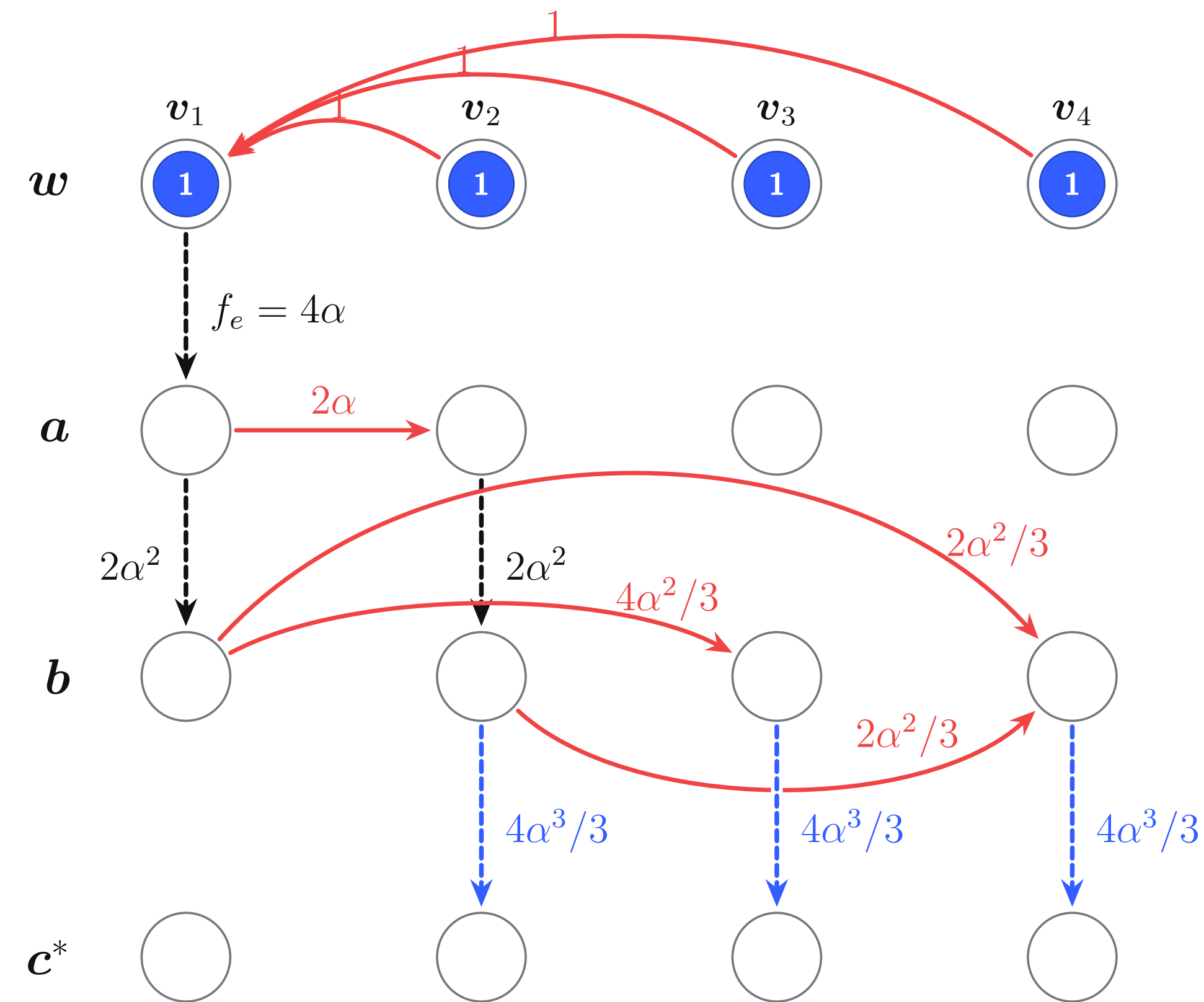
The Analytical Tool: Dual Flow Framework

Graph construction and visual mechanics



The Analytical Tool: Dual Flow Framework

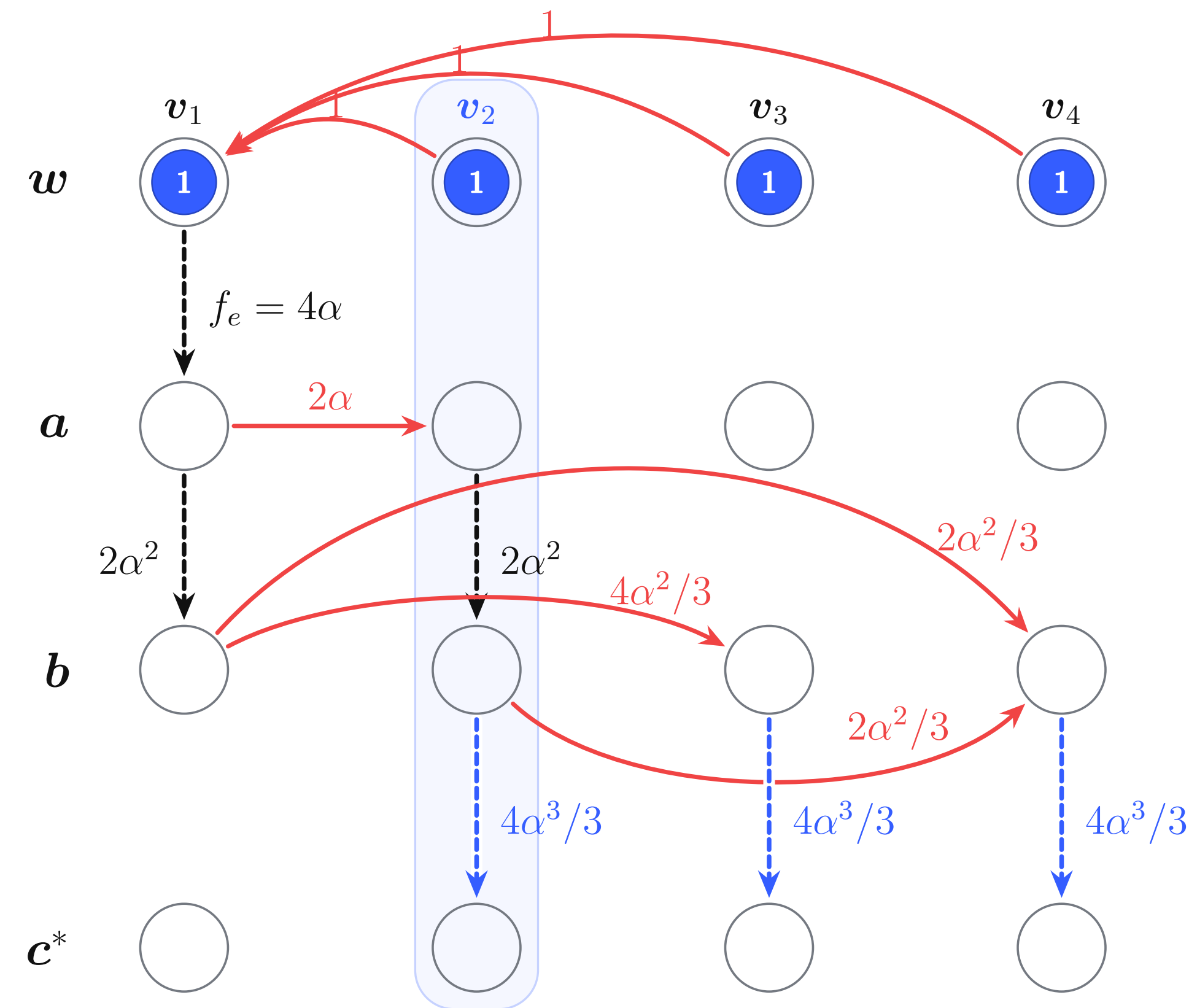
Graph construction and visual mechanics



The Analytical Tool: Dual Flow Framework

Graph construction and visual mechanics

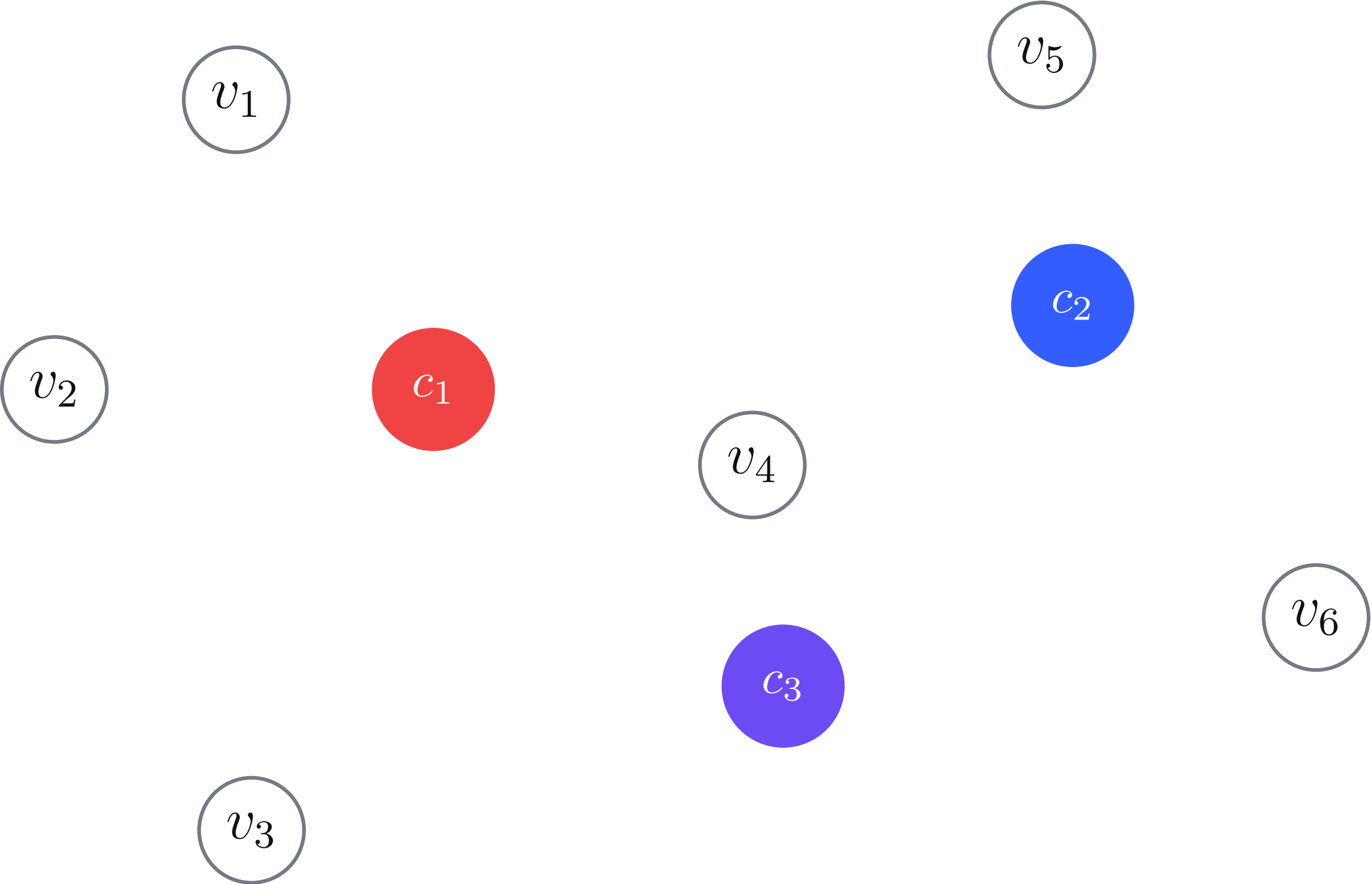
- The cost of f at voter v_2 : $cost(v_2) = 1 + 2\alpha + \frac{2}{3}\alpha^2 + \frac{4}{3}\alpha^3$



- By weak duality, the cost we're minimizing here in the flow upper bounds the maximum distortion

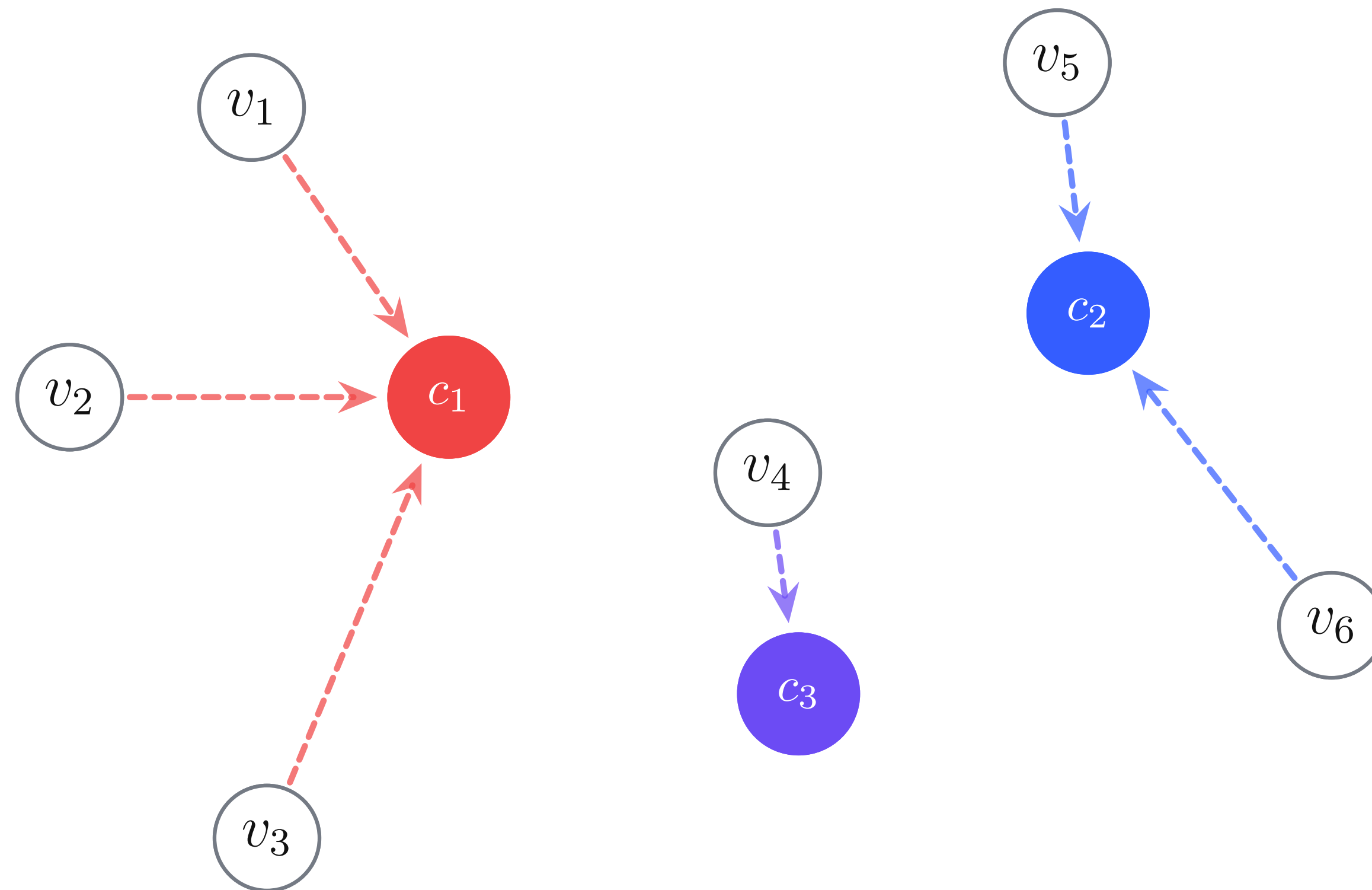
Plurality Upper Bound

$O(\alpha m)$



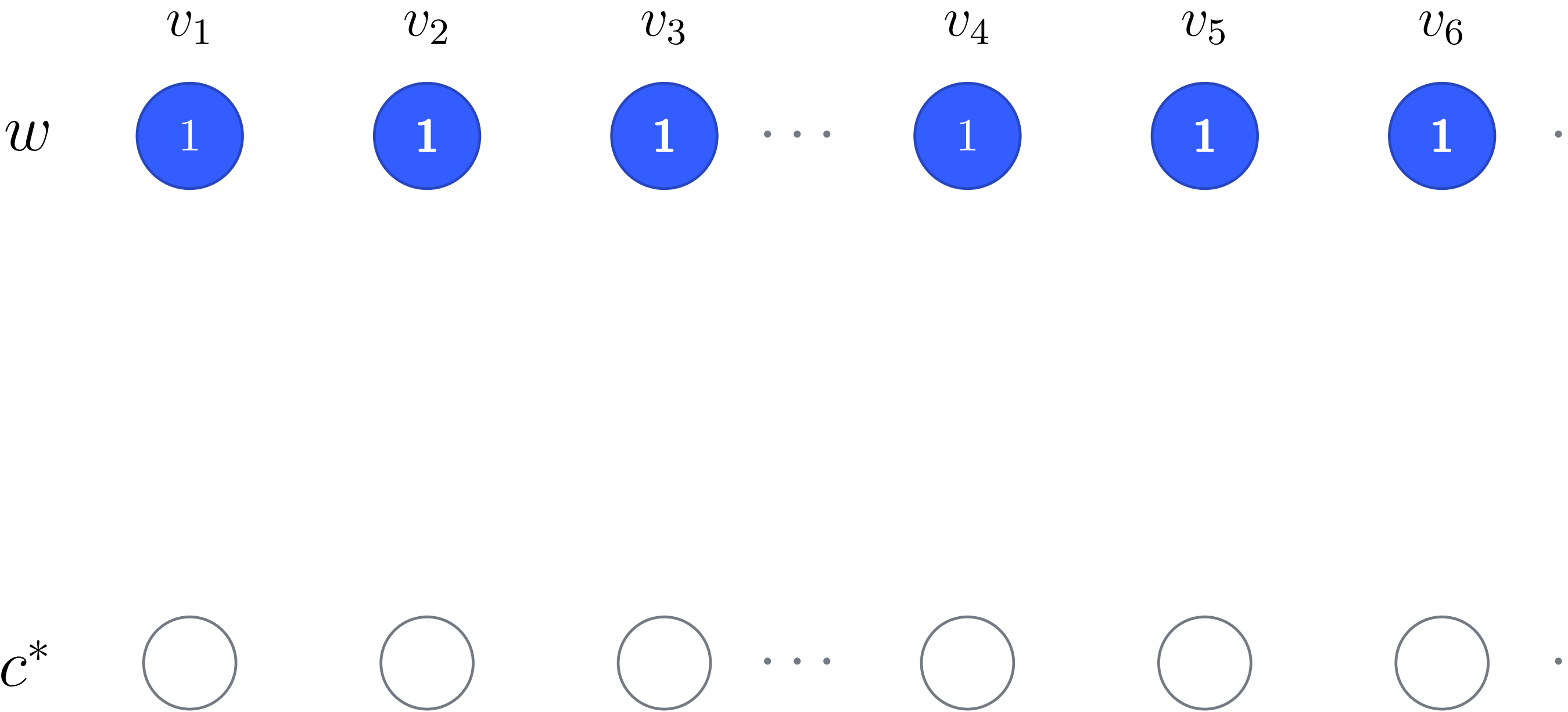
Plurality Upper Bound

$O(\alpha m)$



Plurality Upper Bound

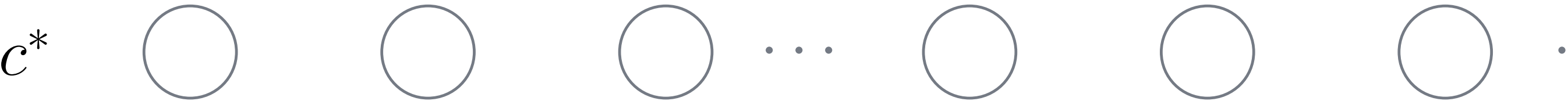
$O(\alpha m)$



$$k = \frac{\text{\#voters}}{\text{\#voters who votes for } w}$$

Plurality Upper Bound

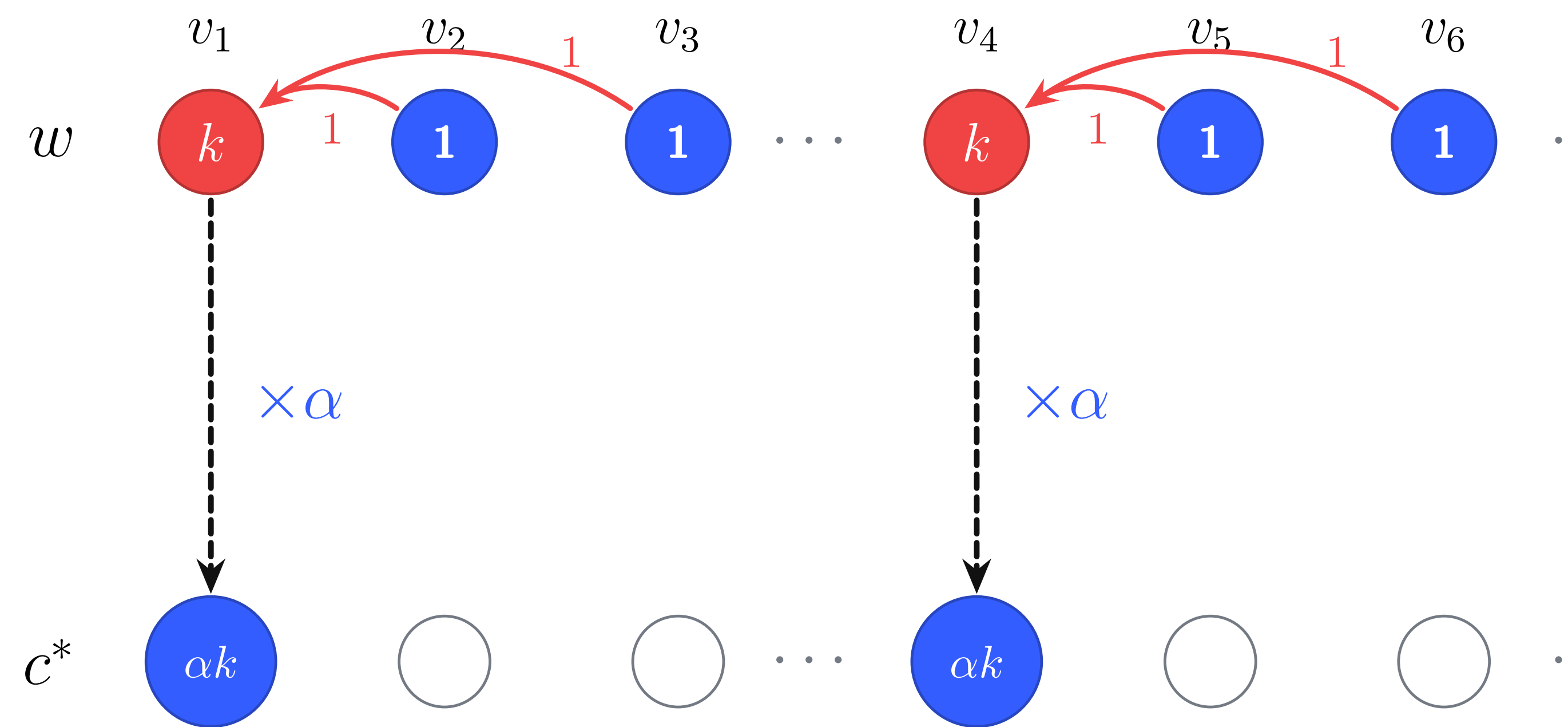
$O(\alpha m)$



$$k = \frac{\text{\#voters}}{\text{\#voters who votes for } w}$$

Plurality Upper Bound

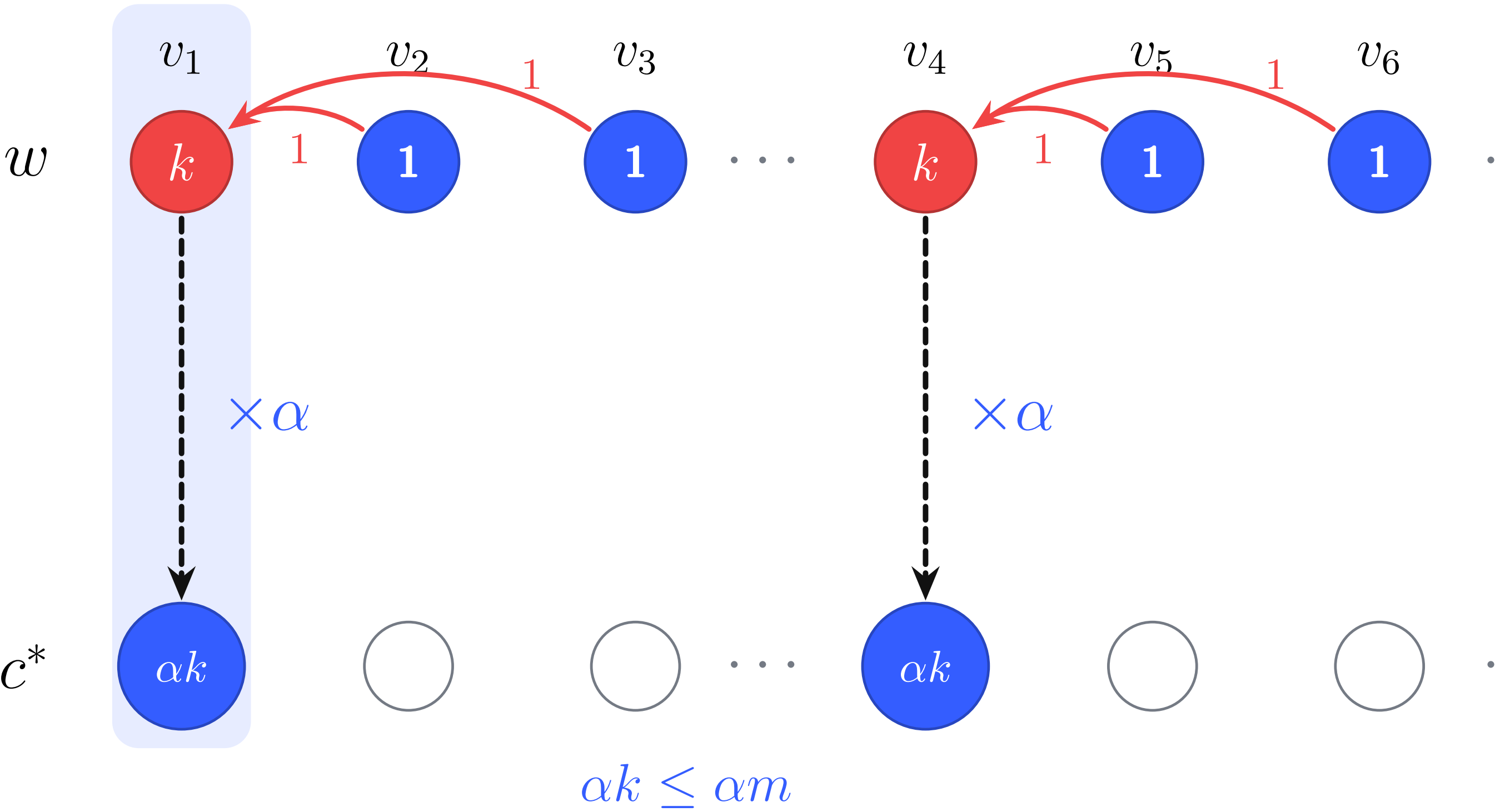
$O(\alpha m)$



$$k = \frac{\text{\#voters}}{\text{\#voters who votes for } w} \leq m$$

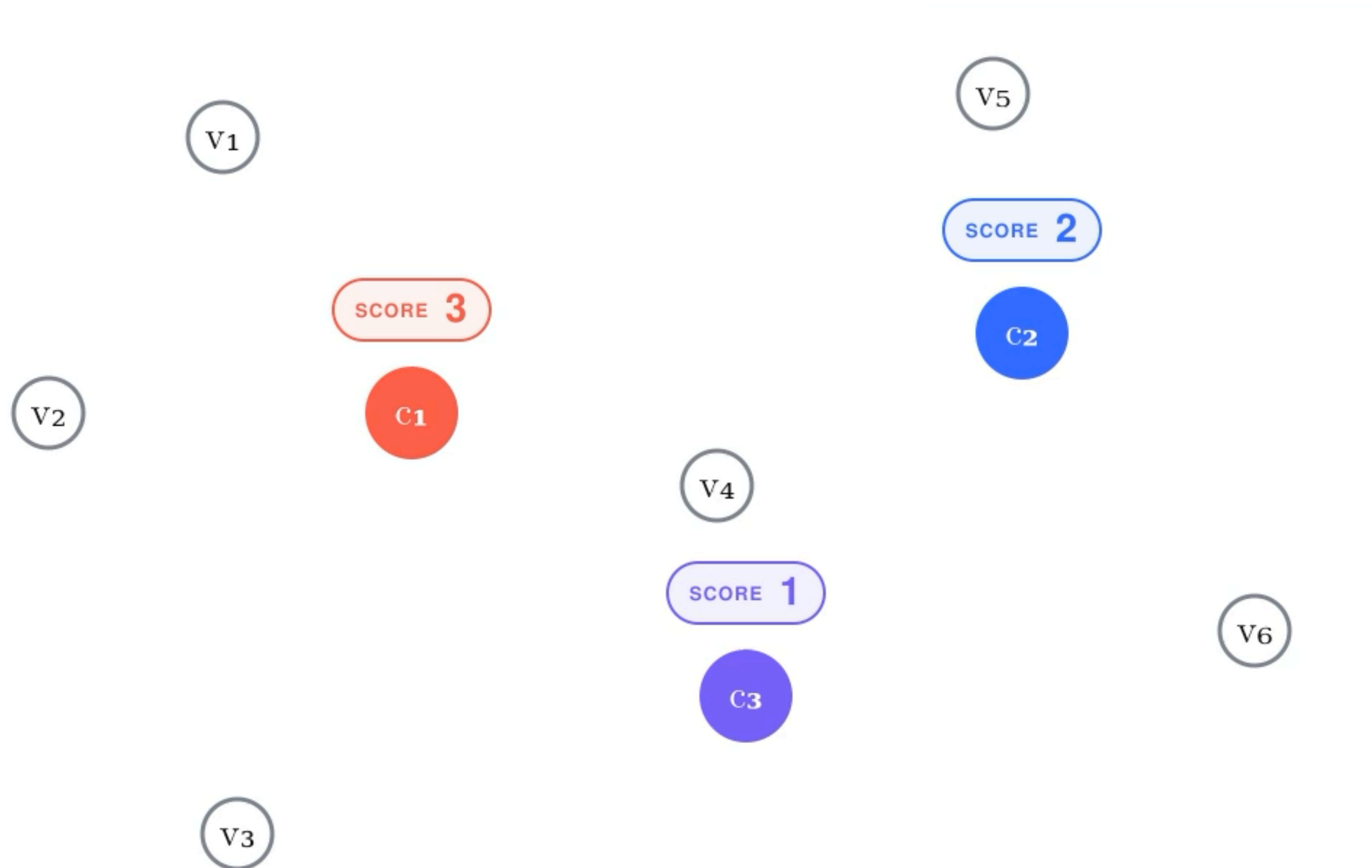
Plurality Upper Bound

$O(\alpha m)$



Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$

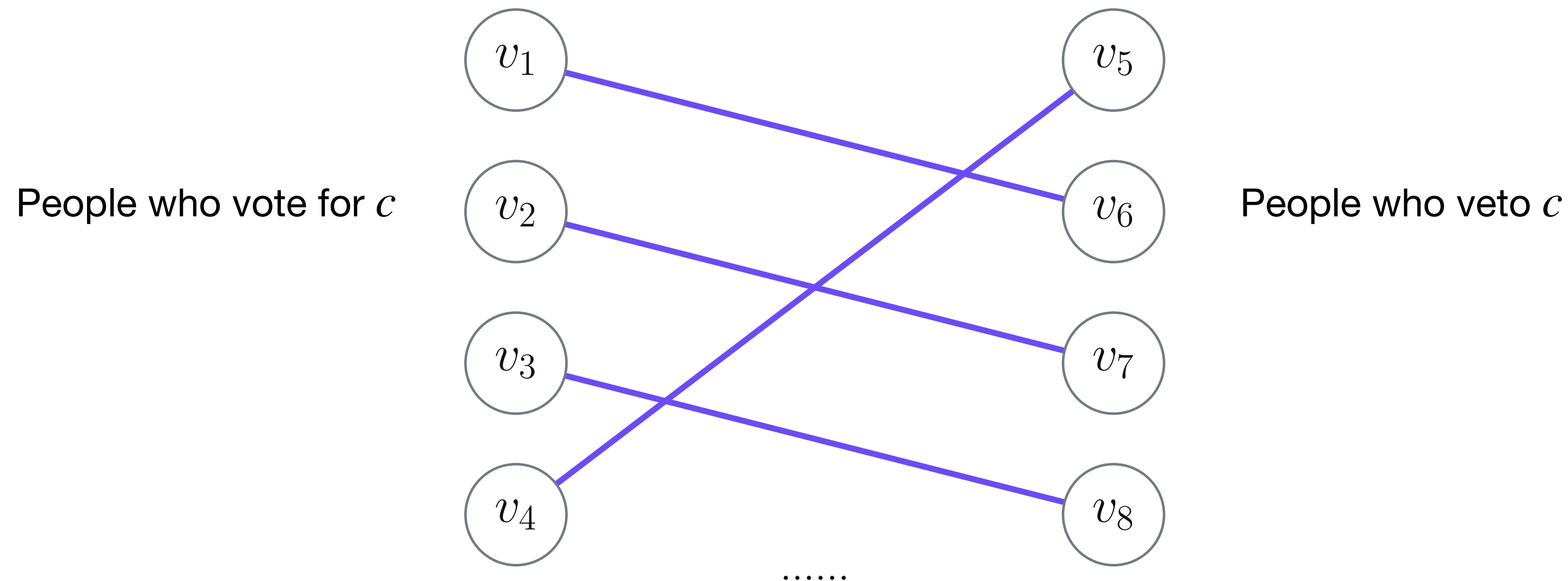


Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$

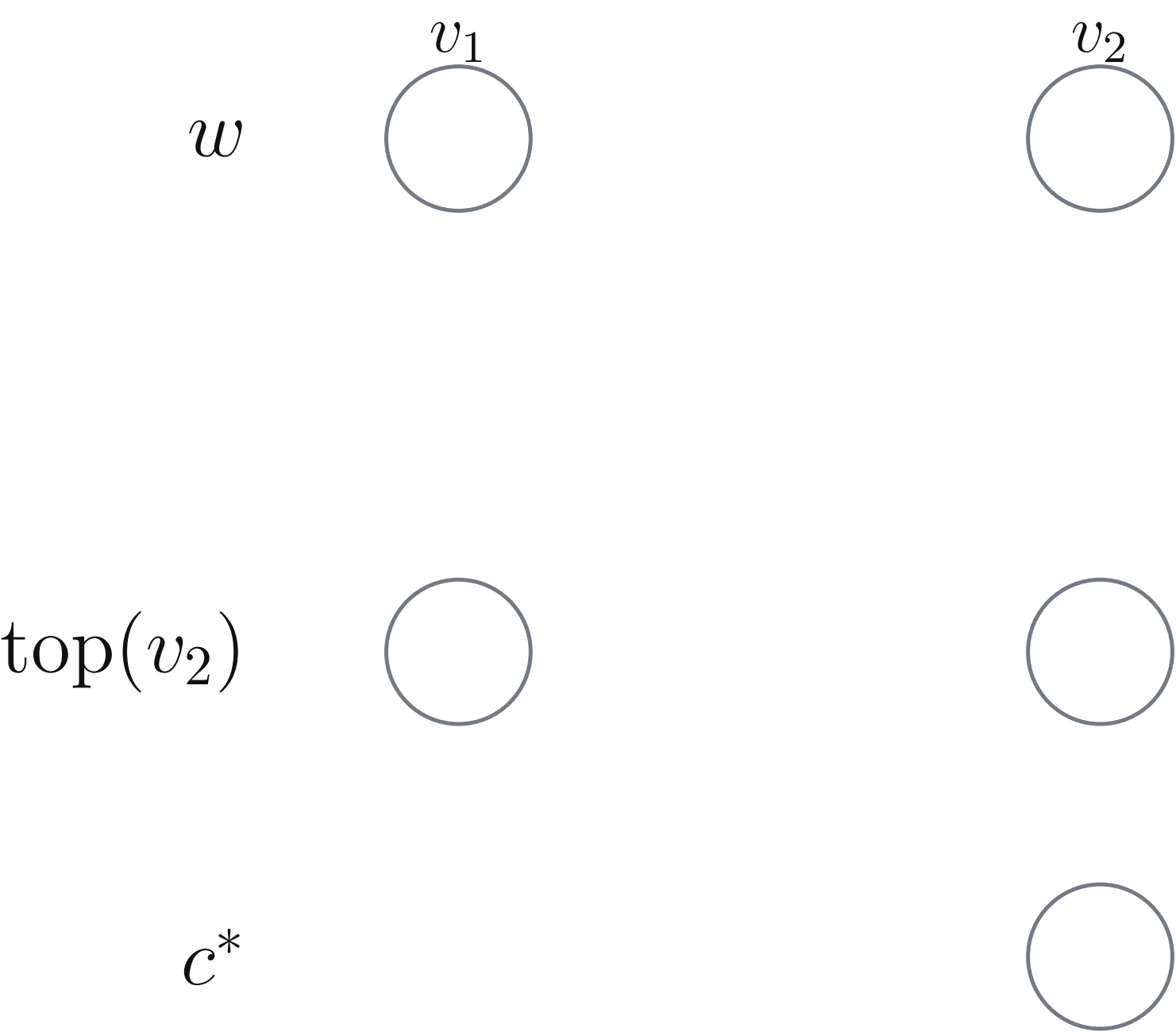
Bipartite matching for voters who vote specifically
for $c \in C \setminus \{w\}$ who does not win

$$V := \{v : \text{top}(v) = c\}$$



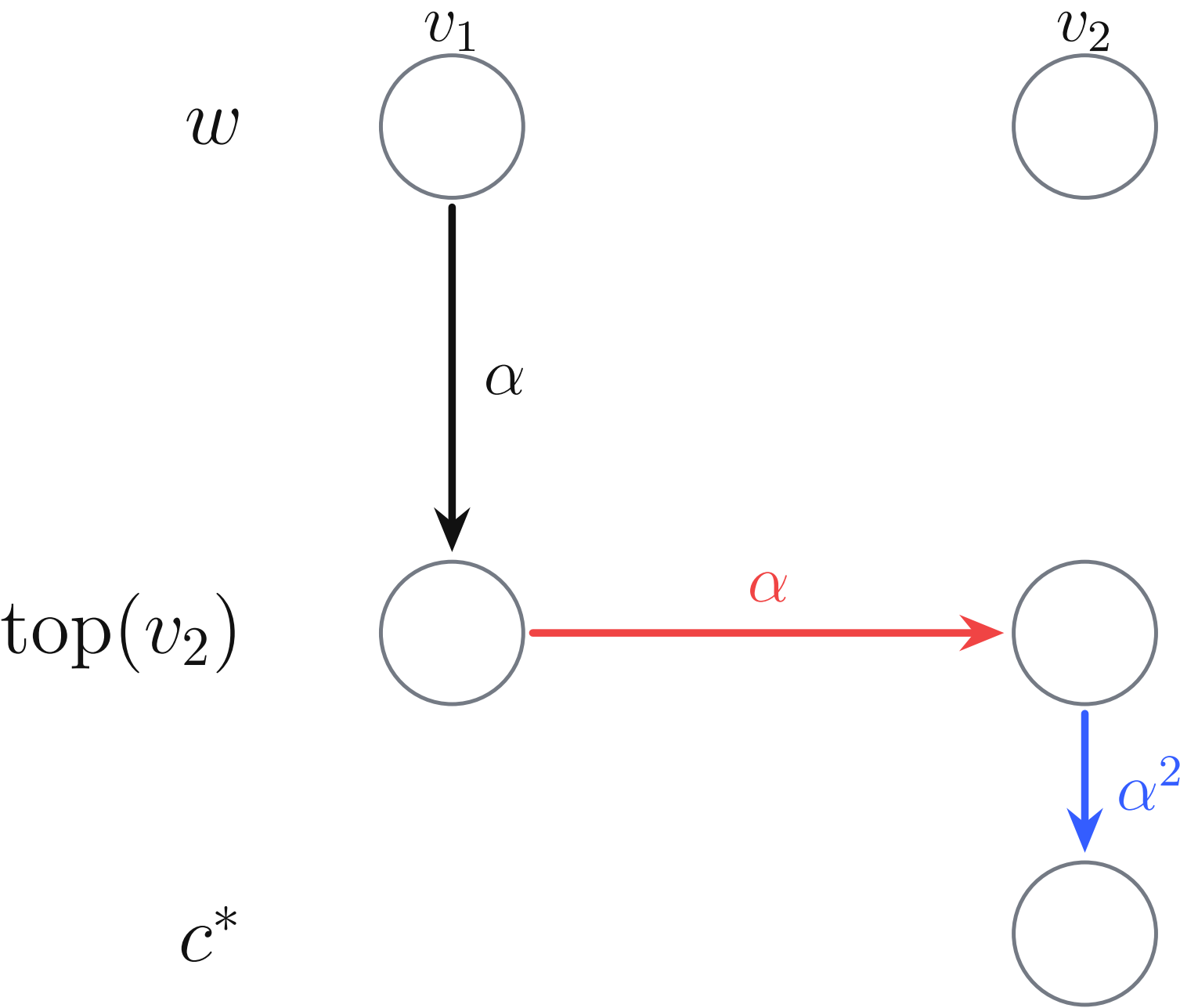
Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$



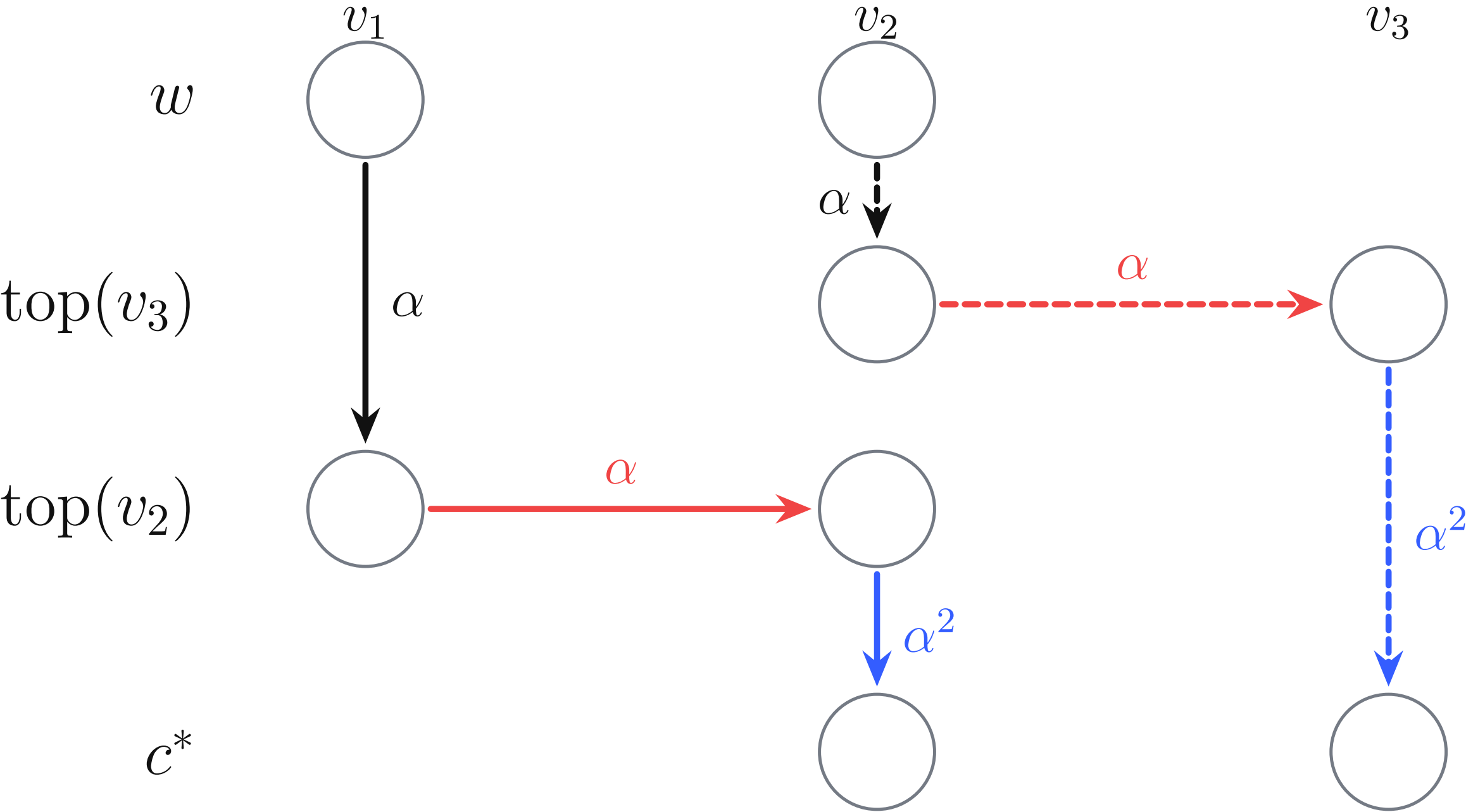
Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$



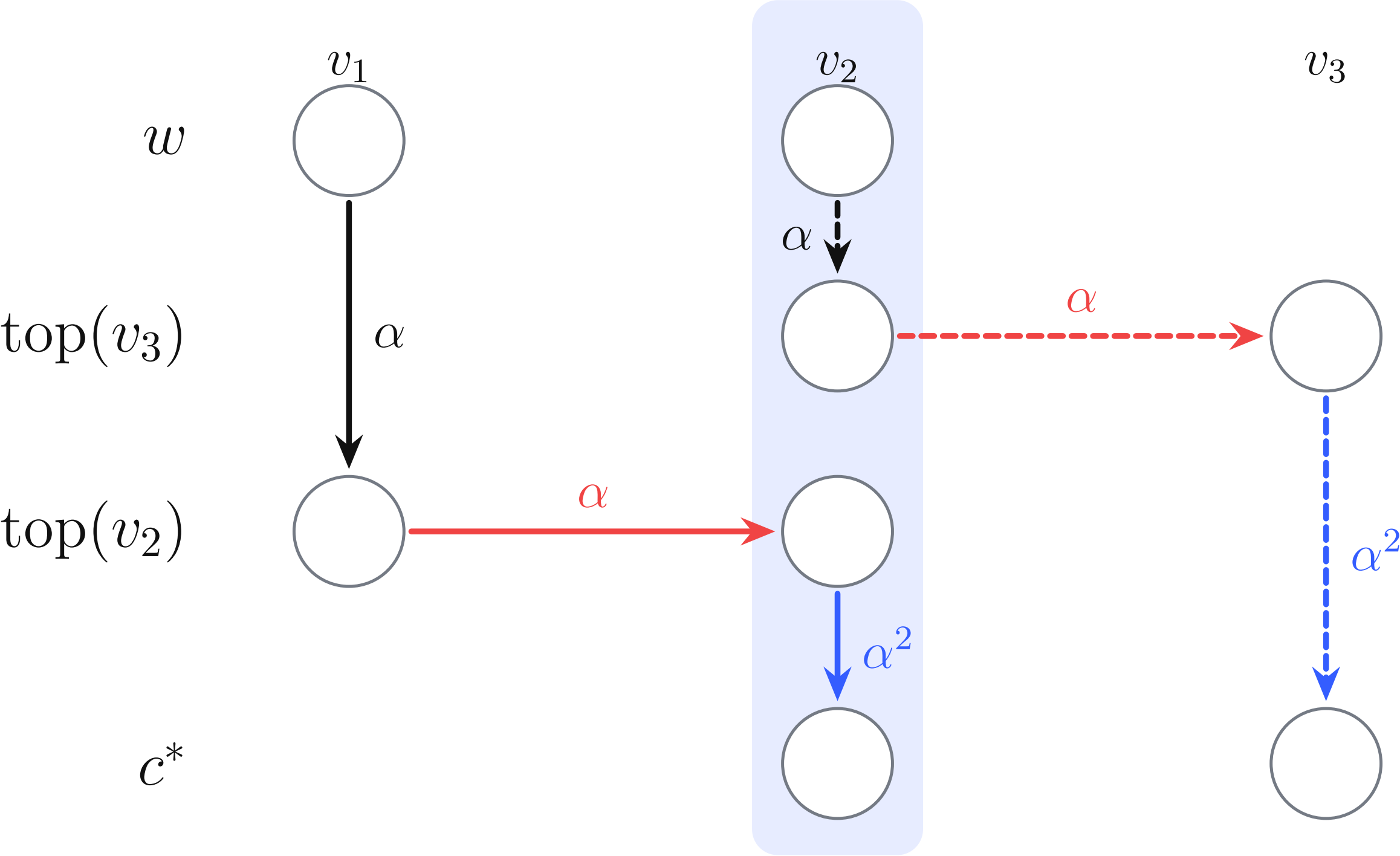
Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$



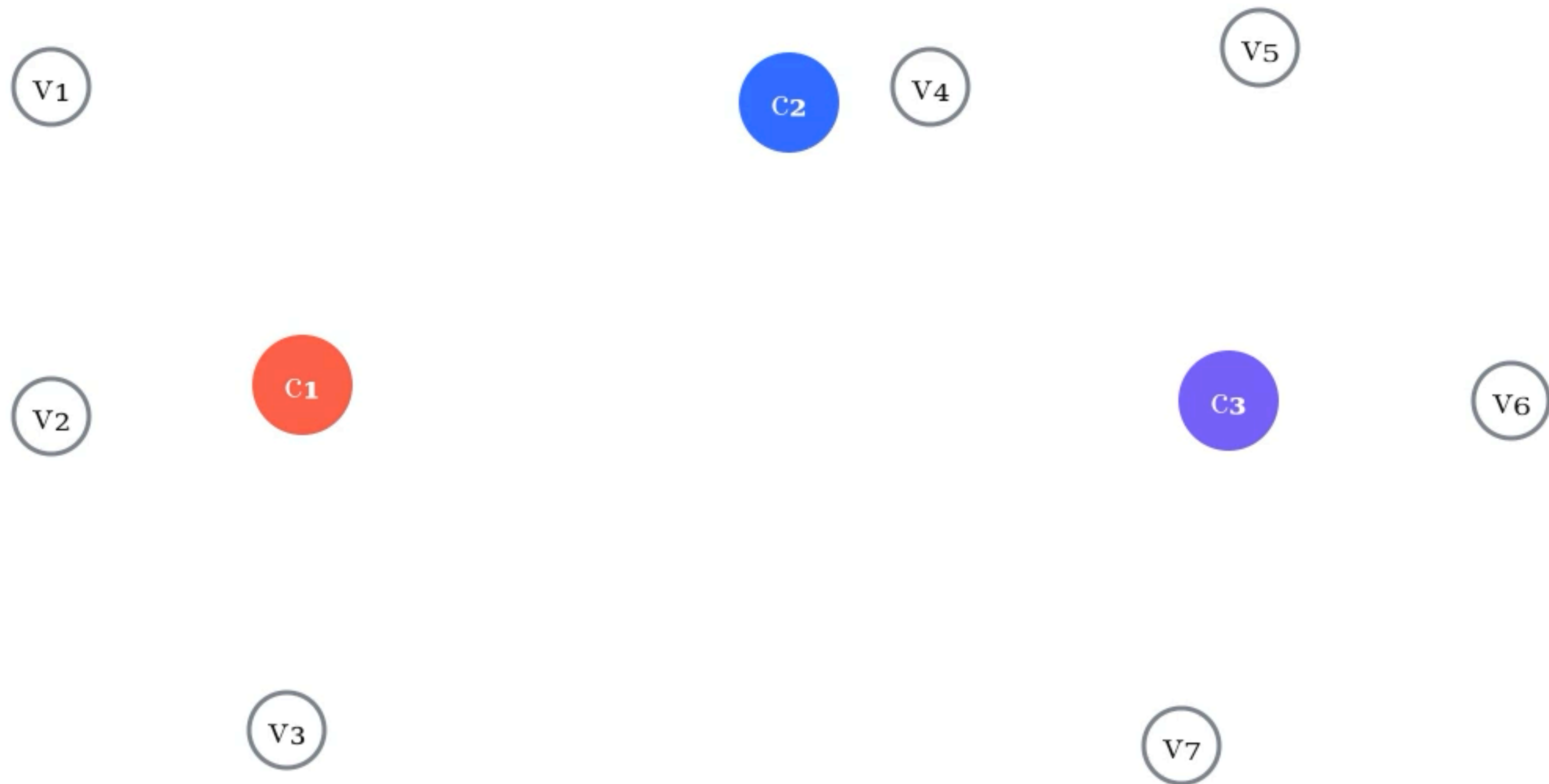
Plurality Veto Upper Bound

$$\alpha^2 + 2\alpha$$



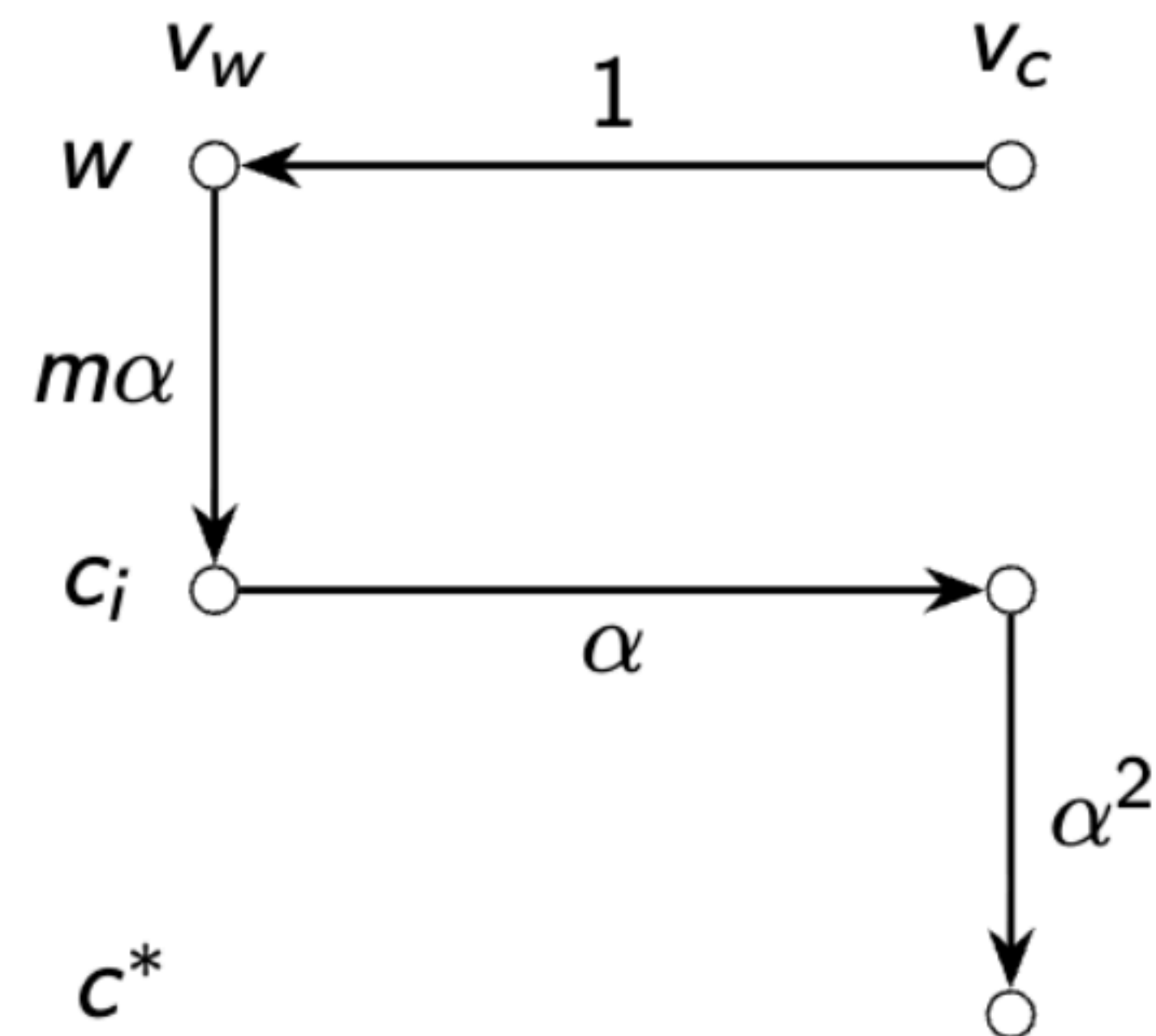
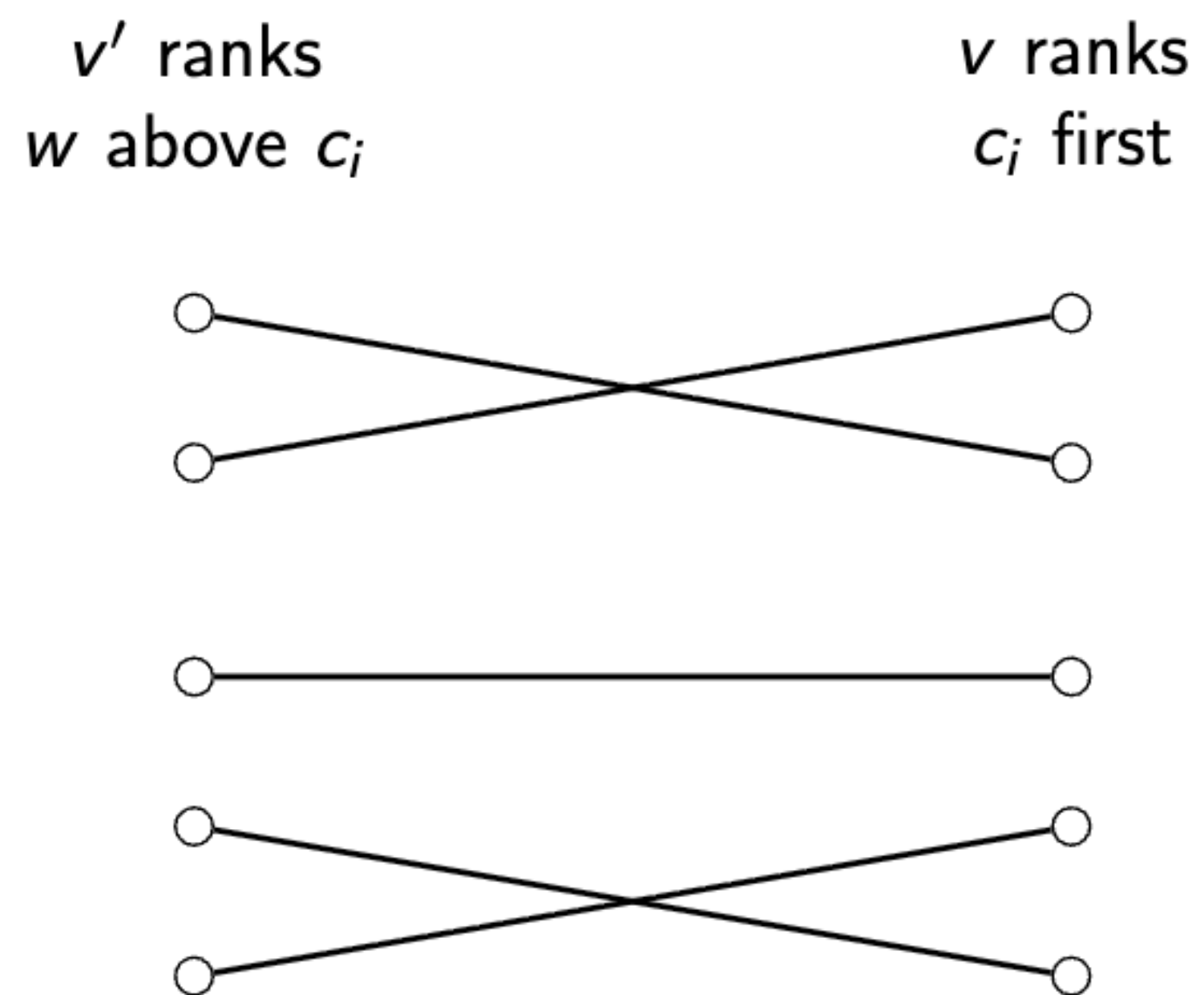
STV Upper Bound

$$O(\max(\alpha m, \alpha^2))$$



STV Upper Bound

$$O(\max(\alpha m, \alpha^2))$$



Conclusion & Future Directions

Key Takeaways

- Rule comparisons change with regime

Plurality and Borda: $\Theta(\alpha m)$

Plurality Veto and Copeland: $\Theta(\alpha^2)$

- When α is small relative to $m \rightarrow$ Plurality Veto and Copeland have the smaller asymptotic bounds
- When α is larger relative to $m \rightarrow$ Plurality and Borda can have the smaller asymptotic bounds

$\Rightarrow$ Rules with poor guarantees in the standard model can become comparatively robust when voters are poorly informed and the number of candidates is small.

Conclusion & Future Directions

Future Work

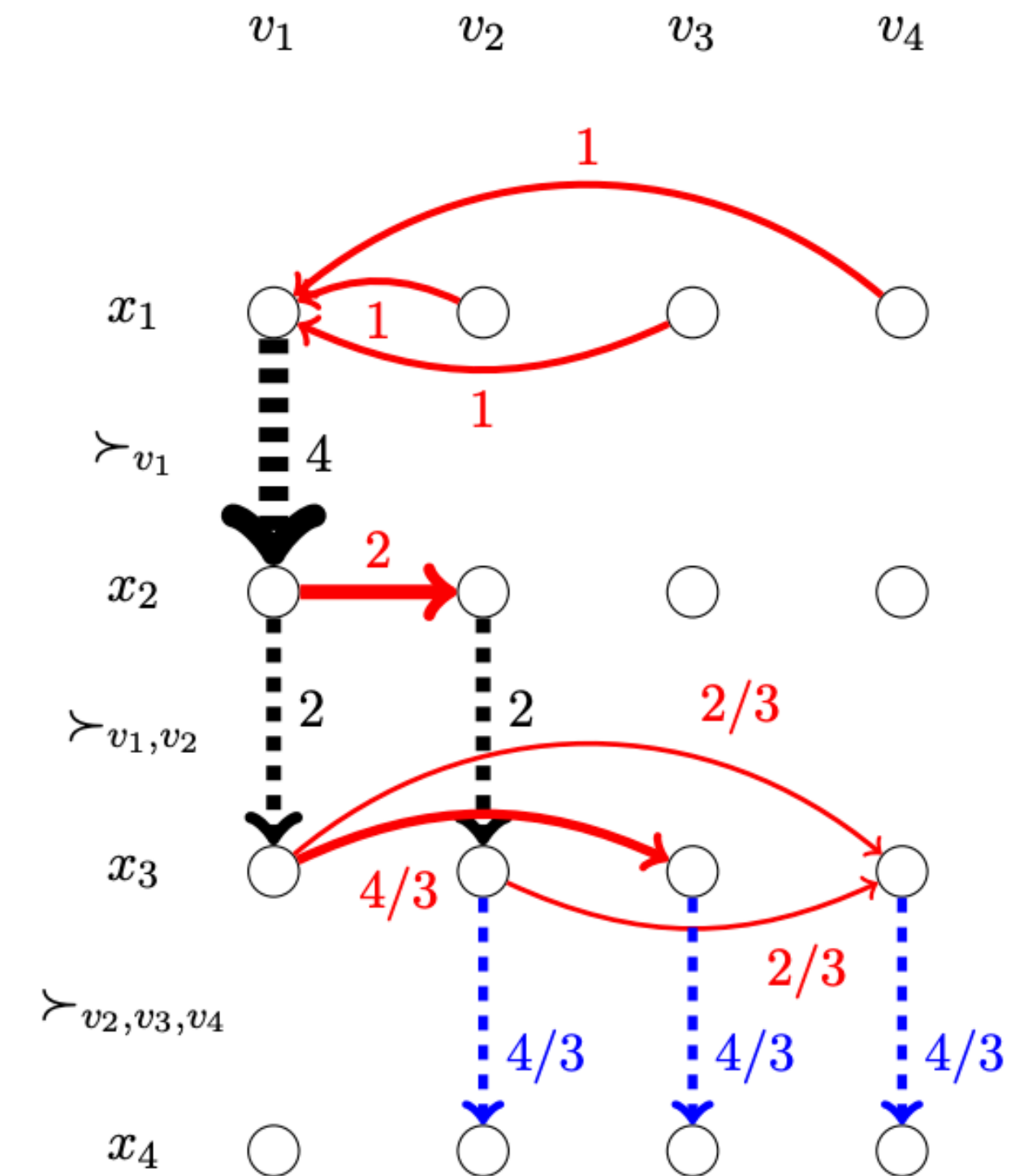
- **Randomized Social Choice:** Extend α -distortion to randomized voting rules to evaluate if probabilistic tie-breaking/selection can mitigate error compounding.
- **Tightening Distortion Bounds:** Resolve remaining gaps between upper and lower bounds for STV and other voting rules.

Appendix

Graph construction and visual mechanics

- $\phi_{x,x'}^{(v)} = f_{(v,x) \rightarrow (v,x')}$ are the vertical preference edges (black and blue)
- $\psi_{x,x'}^{(v,v')} = f_{(v,x) \rightarrow (v',x')}$ are the horizontal sideway edges (red)
- The cost of f at voter v :

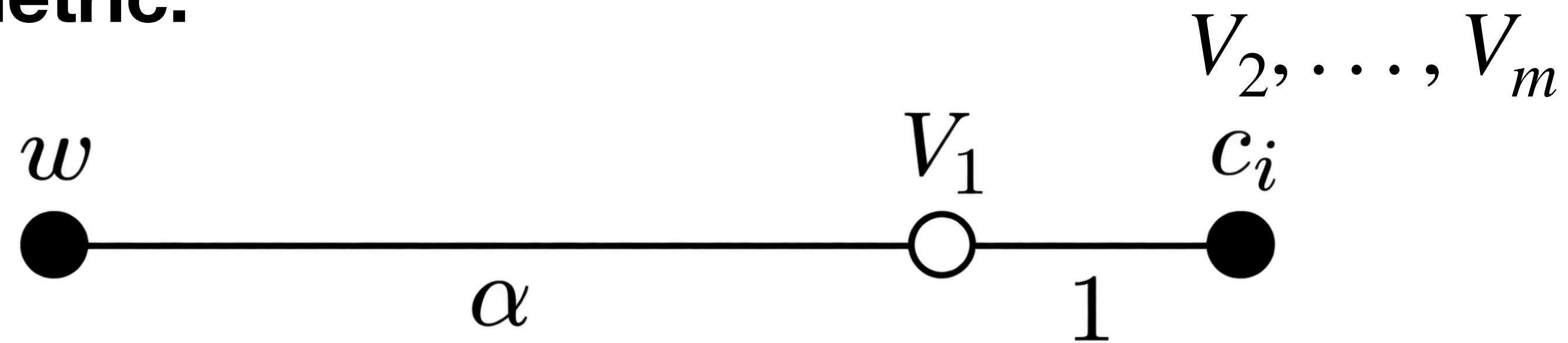
$$\gamma_v^{(f)} = \left(\sum_{e \text{ into } (v,c^*)} \mu_e f_e - \sum_{e \text{ out of } (v,c^*)} f_e \right) + \sum_{c,v'} \left(f_{(v',c) \rightarrow (v,c)} + f_{(v,c) \rightarrow (v',c)} \right)$$
- The cost of f is the maximum cost of f at any voter v : $\gamma(f) = \max_v \gamma_v(f)$



Appendix

Plurality Lower Bound

- Consider the metric:



- And a cyclic preference, where people in V_i rank c_i first.
- w win under plurality rule

- $$\frac{SC(w)}{SC(c^*)} = \frac{\alpha \cdot (k + 1) + (\alpha + 1)(m - 1)k}{(k + 1) \cdot 1}, \text{ equals to } \alpha m \text{ as } k \rightarrow \infty$$

Appendix

Plurality Veto Lower Bound

- **Metric:**

	c_1	c_2	c^*	w
$v \in V_1$	α	$\alpha + 2$	1	$\alpha^2 + 2\alpha$
$v \in V_2$	$\alpha + 2$	α	1	$\alpha^2 + 2\alpha$
v_3, v_4	α^2	α^2	α^2	α^2

- **Ranking:**

$$\begin{array}{l}
 c_1 \succ_{v_1} c^* \succ_{v_1} w \succ_{v_1} c_2 \\
 c_2 \succ_{v_2} c^* \succ_{v_2} w \succ_{v_2} c_1 \\
 w \succ_{v_3} c_1 \succ_{v_3} c_2 \succ_{v_3} c^* \\
 w \succ_{v_4} c_2 \succ_{v_4} c_1 \succ_{v_4} c^*
 \end{array}$$

$$SC(w) = (\alpha^2 + 2\alpha) \cdot 2k + 2\alpha^2$$

$$SC(c^*) = 2k + 2\alpha^2$$

- As $k \rightarrow \infty$, $\frac{SC(w)}{SC(c^*)} \rightarrow \alpha^2 + 2\alpha$